On April 10, 2014, I got an urgent phone call. Chris was having trouble with visualizing the *Endurance*'s trip through the wormhole and he wanted advice. I drove over to his Syncopy compound where postproduction editing was underway, and Chris showed me the problem.

Using my equations, Paul's team had produced visualizations for wormhole trips with various wormhole lengths and lensing widths. For the short, modest-lensing wormhole depicted in the movie, the trip was quick and uninteresting. For a long wormhole, it looked like traveling through a long tunnel with walls whizzing past, too much like things we've seen in movies before. Chris showed me many variants with many bells and whistles, and I had to agree that none had the compelling freshness that he wanted. After sleeping on it, I still had no magic-bullet solution.

The next day Chris flew to London and closeted himself with Paul's Double Negative team, searching for a solution. In the end, they were forced to abandon my wormhole equations and, in Paul's words, "go for a much more abstract interpretation of the wormhole's interior," an interpretation informed by simulations with my equations, but altered significantly to add artistic freshness.

When I experienced the wormhole trip in an early screening of *Interstellar*, I was pleased. Though not fully accurate, it captures the spirit and much of the feel of a real wormhole trip, and it's fresh and compelling.

What did you think?

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28 Recall that Oliver James, chief scientist at Double Negative, wrote the computer code that underlies *Interstellar*'s visualizations of wormholes and black holes; see Chapters 1 and 8.